Higher National Diploma in Information Technology

First Year, Second Semester Examination -2021

HNDIT1214- Statistics for IT

Question 01

(i) Write two benefits of an organization if they use statistics. (4 marks)

(ii) Define qualitative data and quantitative data. (4 marks)

(iii) Following data indicates temperature of patient in Celsius who visited to medical Centre.

37.0	36.9	38.0	37.8	37.7	37.3	36.8
37.0	36.7	37.5	37.2	37.9	37.6	36.7
37.8	35.5	36.2	35.8	35.8	36.1	36.0
37.2	37.2	37.4	37.6	37.3	37.3	37.5
36.7	38.1	38.0	36.6	36.3	35.6	36.6

For the above information:

(a). Find the range (2 marks)

(b). By taking classes as 35.5-35.9, 36.0-36.4, construct a frequency distribution. (6 marks)

(c). If temperature more 37°C is consider as fever, then find the minimum temperature of a fever patient. (3 marks)

(iv) Express following in sigma notation

(a). 1+8+27 +64 + 125

(b). 3+6+11+18+27

(6 marks)

Question 02

Following table represent assignment marks of a group of students.

Marks(class)	No of students(f)
41-50	4
51-60	6
61-70	10
71-80	15
81-90	10
91-100	5

(i) Rearrange the above data in to following table format. (6 marks)

Class Boundaries	Class mark (x)	Frequency (f)	Cumulative Frequency(F)	fx	Fx^2

(ii) Calculate the following measure of central tendency for the above data

(a). Mode (3 marks)

(b). Median (3 marks)

(c). Mean (3 marks)

(iii) Calculate the following measure of dispersion

(a). Inter quartile range (5 marks)

(b). Variance (3 marks)

(c). Standard deviation (2 marks)

Question 03

(i) Briefly explain the followings

(a). Combination

(b). Permutation (6 marks)

(ii) In a stationery shop there are 10 red pen varieties, 12 blue pen varieties and 5 pencil varieties.

(a). How many ways can a person buy one red pen, one blue pen and one pencil from that stationery shop? (3 marks)

(b). If a person wants to buy two red pen and three blue pens, then how many ways can he select this collection? (4 marks)

(iii) A committee of 5 is to be chosen from 4 men and 6 women. In how many ways can this be done if;

(a). There are no restrictions? (3 marks)

(b). The committee consists of women only? (3 marks)

(c). There is at least one man? (3 marks)

(d). There is a majority of women? (3 marks)

Question 04

(i) Define the following terms with an example:

(a). Sample space

(b). Experiment

(c). Outcome

(d). Conditional probability (8 marks)

(ii) The blood groups of 200 people are distributed as follows: 50 have type A blood, 65 have B blood type, 70 have O blood type and 15 have type AB blood. From this group if a person is selected at random,

(a). What is the probability that this person has O blood type? (3 marks)

(b). What is the probability that this person has B blood type or AB blood type? (4 marks)

(iii)In a bag, there are 6 red marbles and 4 green marbles. If a person wants to select two marbles without replacement, then;

(a). Draw a tree diagram to represent above case. (6 marks)

(b). Find the probability that 2 marbles in the same color. (4 marks)

Question 05

(i) Consider the following table

x	0	1	2	3	4	5
P(X=x)	0.10	0.28	0.17	0.11	0.16	0.17

Is it a probability distribution? Prove your answer. (4 marks)

(ii) A random variable has the following probability function.

x	0	1	2	3
P(x)	$\frac{1}{k}$	$\frac{3}{k}$	$\frac{3}{k}$	$\frac{2}{k}$

Find the followings:

(a). The value of k (3 marks)

(b). P ($1 \leq X < 3$) (2 marks)

(c). Expected value (3 marks)

(d). Variance (4 marks)

(iii) The probability of Lahiru being late to school is 0.15. Find the probability that in next 25 mornings he will arrive late to school on;

(a). exactly 6 occasions (3 marks)

(b). less than 6 occasions (3marks)

(c). at least two occasions (3 marks)